Week 08: Programming and Research

Programming task

Session 2: deepen this week’s Python.
Predict → change → repair.
Session 3: apply it to the project.

Research task

In class: discuss Research 6.
No new research assignment.
The project proposal belongs to Programming task.

Open learning_materials/Week_08.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

Programming task

Session 2 · Python practice before project application.

1. Predict JSON → Python

Do this

Open #session-2: Predict → Run.
Decode {"count": 4}, [4], null.
Predict each value and type.

Observe and explain

Object → dict; array → list.
null → None.
Explain which values support .get().

2. Change one field

Do this

Try 4, missing, 0, true and null.
Record .get("count") and its type.
Predict before each run.

Observe and explain

Missing and null both give None here.
true becomes Python True.
Explain why True is not an accepted count.

3. Repair two validation bugs

Do this

Run buggy_valid on 4, true and [].
Copy it as valid_count.
Check dict shape before .get().

Observe and explain

Require a positive exact integer.
Rerun all cases, including missing/null.
Only 4 passes these supplied cases.

Programming practice: done checklist

Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Outside class: complete the practice and save your evidence.

Project application 1: lifecycle report

Step	What to do / check
Open Session 3	#project-task: run supplied fixture/setup; write rescue_metrics(db) and grouped SQL.
Check the report	Offered 94, available 34, reserved 60, collected 42, delivered 29 planned portions; reconcile by request/batch.
Separate buckets	34 available + 18 reserved-only + 13 collected-not-delivered + 29 delivered = 94. Avoid duplicate joins.
Check and explain	Inspect supplied event checks; explain unchanged stock. Keep one calculation trace and limitation.

Use EXISTS or grouped events. Containers, planned portions and people are different measures.

Project application 2: queue, HTTP and proposal

Step	What to do / check
Build the queue	#project-queue-task: six entries with ID, source, outcome, reasons and next action; label offline simulation.
Test the inputs	Boolean count, non-object candidate, cancellation contradiction and M6 outage; compare full DB snapshots.
Run actual HTTP	Enable RUN_LIVE_HTTP. Use a fresh service block. Save six HTTP outcomes, including M6 503.
Save and propose	Assert unchanged DB. Save report/queues, one user need, two cases and a human decision.

If HTTP is blocked, record the real error and arrange a classroom run; never label simulation as HTTP.